



Smart Contract Code Review And Security Analysis Report

CUSTOMER Aryze
SCOPE RYZE Token Contract
DATE 24.02.2026

Made in Germany by softstack GmbH

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1. Disclaimer

The audit makes no statements or warranties about utility of the code, safety of the code, suitability of the business model, investment advice, endorsement of the platform or its products, regulatory regime for the business model, or any other statements about fitness of the contracts to purpose, or their bug free status. The audit documentation is for discussion purposes only.

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Version / Date	Description
0.1 (20.02.2026)	Layout
0.5 (22.02.2026)	Manual + Automated Security Testing
1.0 (24.02.2026)	Final document

2. About the Project and Company

Company: ARYZE ApS

Address: Bredgade 6, 3 København K 1260, Denmark

Website: <https://www.aryze.io>

Twitter (X): <https://x.com/ARYZEofficial>

LinkedIn: <https://www.linkedin.com/company/aryzeofficial/>

ARYZE is a Denmark-based fintech company building fully collateralized stablecoin infrastructure for businesses, governments, and institutions. Their platform offers Stablecoin-as-a-Service, enabling clients to launch branded stablecoins, facilitate instant cross-border payments, and tokenize real-world assets such as fiat currencies and gold.

The **\$RYZE** token is the utility token powering the ARYZE stablecoin ecosystem. It serves as the liquidity unifier across the platform, enabling yield capture and ecosystem participation. The token is implemented as an ERC-20 upgradeable contract with built-in whitelist controls and integrates with the Reforge bridge system for cross-chain token migration.

The audited scope covers the RYZE token contract (`ARYZE_RYZE.sol`), the whitelist access control module (`WhitelistUpgradeable.sol`), and the Reforge V3 router contracts that handle cross-chain bridging and token reforging operations.

3. Vulnerability & Risk Level

Risk represents the probability that a certain source-threat will exploit vulnerability, and the impact of that event on the organization or system. Risk Level is computed based on CVSS version 3.1.

Level	CVSS Score	Vulnerability	Required Action
CRITICAL	9.0 – 10.0	A vulnerability that can disrupt the contract functioning in a number of scenarios, or creates a risk that the contract may be broken.	Immediate action to reduce risk level.
HIGH	7.0 – 8.9	A vulnerability that affects the desired outcome when using a contract, or provides the opportunity to use a contract in an unintended way.	Implementation of corrective actions as soon as possible.
MEDIUM	4.0 – 6.9	A vulnerability that could affect the desired outcome of executing the contract in a specific scenario.	Implementation of corrective actions in a certain period.
LOW	0.1 – 3.9	A vulnerability that does not have a significant impact on possible scenarios for the use of the contract and is probably subjective.	Implementation of certain corrective actions or accepting the risk.
INFO	0.0	A vulnerability that has informational character but is not affecting any of the code.	An observation that does not determine a level of risk.

4. Auditing Strategy and Techniques Applied

Throughout the review process, care was taken to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices.

The auditing process follows a routine series of steps:

Code review that includes the following:

- Review of the specifications, sources, and instructions provided to softstack to make sure we understand the size, scope, and functionality of the smart contract.
- Manual review of code, which is the process of reading source code line-by-line in an attempt to identify potential vulnerabilities.
- Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to softstack describe.

Testing and automated analysis that includes the following:

- Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
- Symbolic execution, which is analysing a program to determine what inputs causes each part of a program to execute.

Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarify, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.

Specific, itemized, actionable recommendations to help you take steps to secure your smart contracts.

5. Metrics

The metrics section gives the reader an overview of the size, quality, flows and capabilities of the codebase.

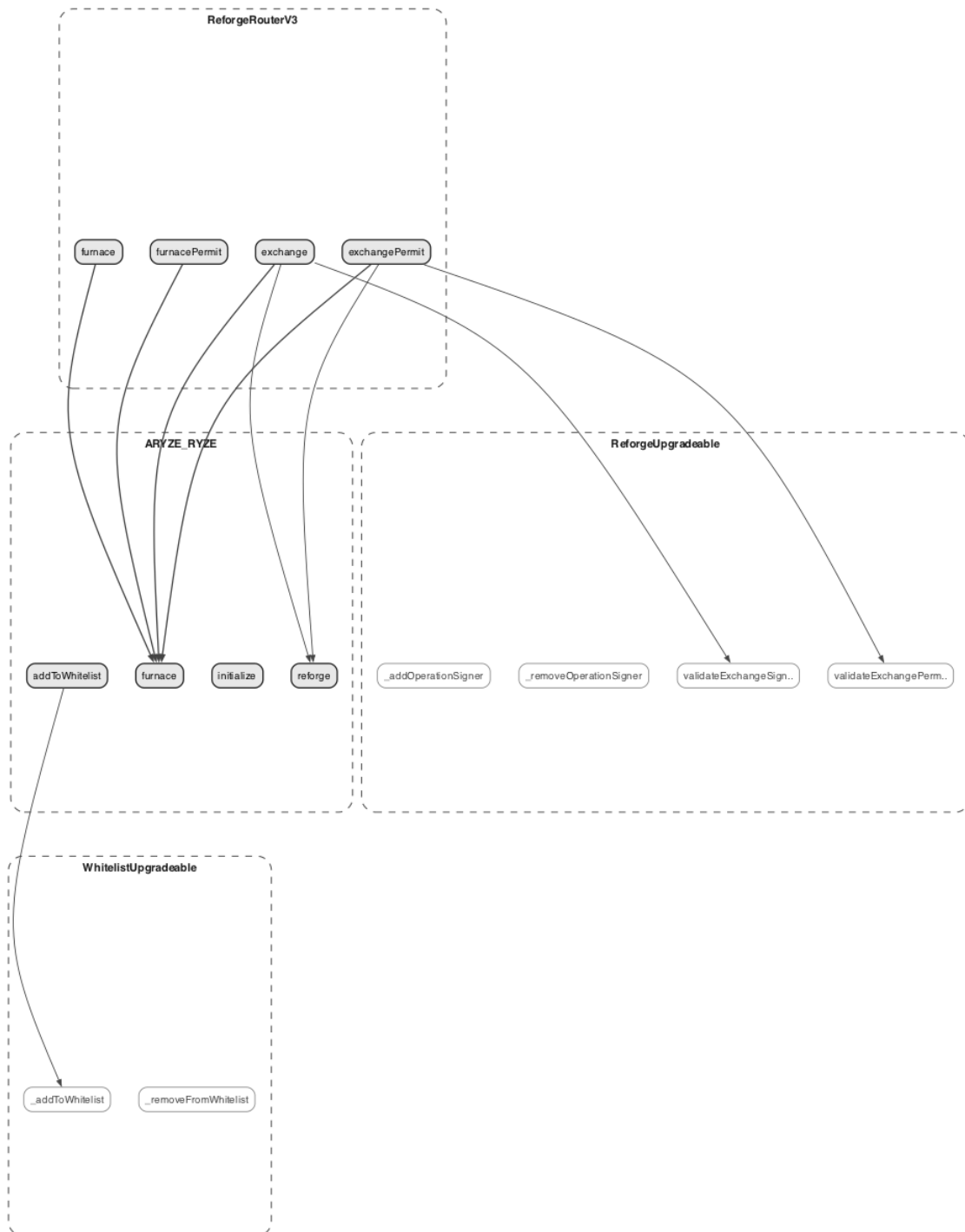
5.1 Tested Contract Files

Source: <https://github.com/ARYZEofficial/aryze-ryze>

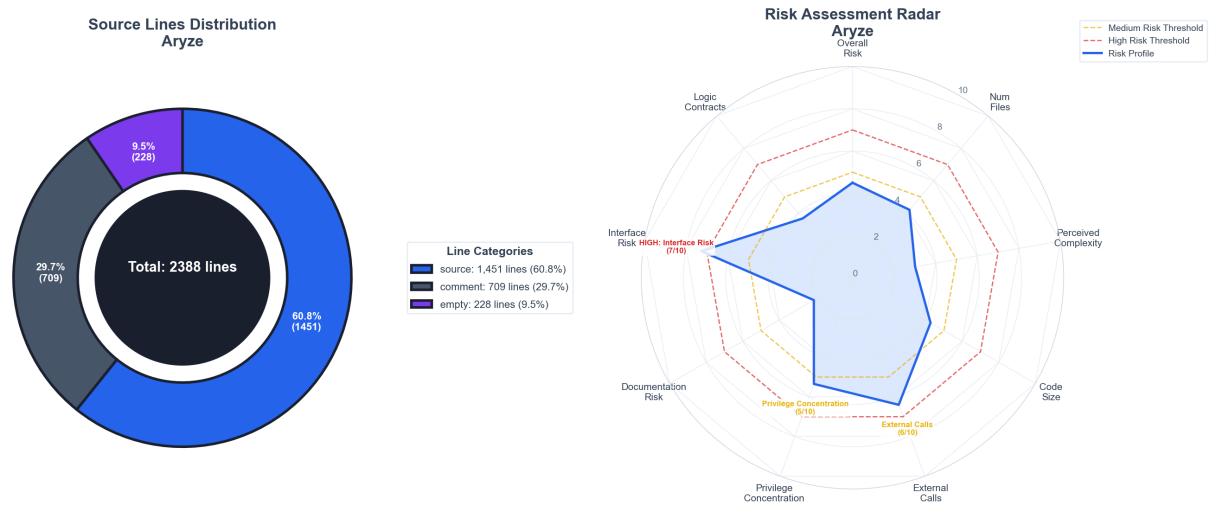
Commit: 4dd03aba3b25e9715f34f8466d0140a33acffd0e

File	Fingerprint (MD5)
./aryze_ryze/contracts/new_chain_reforge_steps/3.tokens/bridge_v2_LATEST/RYZE/ARYZE_RYZE.sol	f77eca44c9fbe5c26da69134791b3fc1
./aryze_ryze/contracts/new_chain_reforge_steps/3.tokens/bridge_v2_LATEST/RYZE/WhitelistUpgradeable.sol	a7b52bf93b94f54bdf39ff5021c7be31
./aryze_ryze/contracts/new_chain_reforge_steps/3.tokens/bridge_v2_LATEST/RYZE/interfaces/IMinecartEvents.sol	d38566f51326a5397ae3aa323b119cc6
./aryze_ryze/contracts/new_chain_reforge_steps/3.tokens/bridge_v2_LATEST/RYZE/interfaces/IReforgeUnified.sol	b6efcaca3237906fc49cf28cfad6eee8
./reforge-v3/ReforgeRouter/ReforgeRouterV3.sol	cb2c7bb2c1f14ddfeefdb3c49e0c71e8
./reforge-v3/ReforgeRouter/core/ReforgeUpgradeable.sol	1608230523f5c290c3fa9e6b52cae5c9
./reforge-v3/ReforgeRouter/interfaces/IMinecartEvents.sol	3a96ab55ae9cc385f305dacba7766ede
./reforge-v3/ReforgeRouter/interfaces/INTReforgeUnified.sol	a2d06f579387f0a3141680081813c0f1
./reforge-v3/ReforgeRouter/interfaces/IReforgeUnified.sol	d47333535ca500b5b5a01a6f84057fe3
./reforge-v3/ReforgeRouter/interfaces/ISignature.sol	44e222b14000c308e2ca6e960f13a950
./reforge-v3/ReforgeRouter/interfaces/IValidationMemory.sol	d6bcd654b8739eef521c0339a3be533f

5.2 Call Graph



5.3 Source Lines & Risk



5.4 Capabilities

Module	Functions	Public	Restricted	Key Roles
ARYZE_RYZE	28	9	19	DEFAULT_ADMIN, BRIDGE_OWNER, PAUSER
INTReforgeUnified	3	3	0	-
IReforgeUnified	6	6	0	-
ReforgeRouterV3	20	9	11	DEFAULT_ADMIN, PAUSER, UPGRADER
ReforgeUpgradeable	11	11	0	-
WhitelistUpgradeable	1	1	0	-

5.5 Dependencies / External Imports

Import Path	Version
@openzeppelin/contracts-upgradeable/access/AccessControlUpgradeable.sol	v5.0.0
@openzeppelin/contracts-upgradeable/proxy/utils/Initializable.sol	v5.0.0
@openzeppelin/contracts-upgradeable/proxy/utils/UUPSUpgradeable.sol	v5.0.0
@openzeppelin/contracts-upgradeable/utils/ReentrancyGuardUpgradeable.sol	v5.0.0
@openzeppelin/contracts-upgradeable/utils/cryptography/EIP712Upgradeable.sol	v5.0.0

Import Path	Version
@openzeppelin/contracts/access/IAccessControl.sol	v5.0.0
@openzeppelin/contracts/interfaces/IERC5267.sol	v5.0.0
@openzeppelin/contracts/interfaces/draft-IERC1822.sol	v5.0.0
@openzeppelin/contracts/interfaces/draft-IERC6093.sol	v5.0.0
@openzeppelin/contracts/proxy/ERC1967/ERC1967Utils.sol	v5.0.0
@openzeppelin/contracts/token/ERC20/IERC20.sol	v5.0.0
@openzeppelin/contracts/token/ERC20/extensions/IERC20Metadata.sol	v5.0.0
@openzeppelin/contracts/token/ERC20/extensions/IERC20Permit.sol	v5.0.0
@openzeppelin/contracts/utils/Address.sol	v5.0.0
@openzeppelin/contracts/utils/cryptography/ECDSA.sol	v5.0.0
@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol	v5.0.0
@openzeppelin/contracts/utils/introspection/IERC165.sol	v5.0.0

5.6 Source Units in Scope

File	Instructions	Functions	Lines	nSLOC	Comments
ReforgeRouterV3.sol	19	20	652	406	194
ReforgeUpgradeable.sol	11	41	1088	750	232
IValidationMemory.sol	0	0	21	9	11
IReforgeUnified.sol	3	3	16	7	5
INTReforgeUnified.sol	3	3	26	7	15
ISignature.sol	0	0	25	12	12
IMinecartEvents.sol	0	0	14	5	7
WhitelistUpgradeable.sol	1	3	41	18	17
ARYZE_RYZE.sol	24	29	463	225	192
IReforgeUnified.sol	3	3	28	7	17
IMinecartEvents.sol	0	0	14	5	7
Total	64	102	2388	1451	709

6. Scope of Work

The ARYZE team has provided the Solidity smart contracts for their RYZE utility token and Reforge bridge system. The audit focuses on validating security, access control, upgradeability patterns, and correctness of the token logic, whitelist management, and cross-chain bridge mechanisms.

6.1 Findings Overview

Status: No findings were identified during the security assessment.

6.2 Manual and Automated Vulnerability Test

No findings were identified during the manual and automated security testing of the smart contracts in scope.

7. Executive Summary

Two independent softstack experts performed an unbiased and isolated audit of the smart contract provided by the Aryze team. The main objective of the audit was to verify the security and functionality claims of the smart contract. The audit process involved a thorough manual code review and automated security testing.

Overall, the audit identified a total of 0 issues, classified as follows:

- No critical issues were found
- No high issues were found
- No medium issues were found
- No low issues were found
- No informational issues were found

The audited codebase comprises 1,451 normalized source lines of code across 11 files.

\$RYZE Token Deployments

Network	Contract Address	Explorer
Ethereum	0x44e3ae622c1570dc6e492adb8de92d01ca923d26	View on Explorer
Polygon	0x44e3ae622c1570dc6e492adb8de92d01ca923d26	View on Explorer
Linea	0x34b5c2936dae6698e54781edb8b1e69a2c2873f8	View on Explorer
BNB Chain	0x7712da72127d5dd213b621497d6e4899d5989e5c	View on Explorer
Arbitrum	0x7712da72127d5dd213b621497d6e4899d5989e5c	View on Explorer

8. About the Auditor

Established in 2017 under the name Chainsulting, and rebranded as softstack GmbH in 2023, softstack has been a trusted name in Web3 Security space. Within the rapidly growing Web3 industry, softstack provides a comprehensive range of offerings that include software development, cybersecurity, and consulting services. Softstack's competency extends across the security landscape of prominent blockchains like Solana, Tezos, TON, Ethereum and Polygon. The company is widely recognized for conducting thorough code audits aimed at mitigating risk and promoting transparency.

The firm's proficiency lies particularly in assessing and fortifying smart contracts of leading DeFi projects, a testament to their commitment to maintaining the integrity of these innovative financial platforms. To date, softstack plays a crucial role in safeguarding over \$100 billion worth of user funds in various DeFi protocols.

Underpinned by a team of industry veterans possessing robust technical knowledge in the Web3 domain, softstack offers industry-leading smart contract audit services. Committed to evolving with their clients' ever-changing business needs, softstack's approach is as dynamic and innovative as the industry it serves.

Check our website for further information: <https://softstack.io>

9. Glossary

Term	Definition
CVSS	Common Vulnerability Scoring System
DeFi	Decentralized Finance
ERC	Ethereum Request for Comments
ERC-20	Ethereum token standard for fungible tokens
OZ	OpenZeppelin, widely-used Solidity library for secure smart contracts
RYZE	Utility token powering the ARYZE stablecoin ecosystem
Reforge	Cross-chain bridge and token migration system
UUPS	Universal Upgradeable Proxy Standard